Tyrosine kinase inhibitors (TKI)

Tyrosine kinase inhibitors (TKIs) are a type of targeted therapy. TKIs are the main treatment for chronic myeloid leukaemia (CML).

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Treating chronic myeloid leukaemia (CML) with TKIs

The main treatment for chronic myeloid leukaemia (CML) is drugs called tyrosine kinase inhibitors (TKIs). TKIs are a type of targeted therapy.

The TKI drugs used to treat CML come as tablets or capsules. You usually take them every day for as long as they are working.

How do TKIs work?

TKIs work by by switching off (inhibiting) the tyrosine kinase that is made by the BCR-ABL1 gene in the leukaemia cells. This stops the bone marrow from making abnormal white blood cells, or from making so many. It also allows the leukaemia cells to mature and die.

Types of TKIs

The 3 main TKI drugs currently used in the UK for people newly diagnosed with CML are:

Imatinib (Glivec®)

Imatinib is the most commonly used TKI for CML. It can be used in any phase.

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Nilotinib (Tasigna®)

Nilotinib can be used as a first treatment in the chronic phase. It can also be used in any other phase if other TKIs are causing severe side effects or are not working to control the CML.

Dasatinib (Sprycel®)

Dasatinib can be used as a first treatment in the chronic phase. It can also be used in any phase if other treatments are causing severe side effects or is not working to control the CML.

Other less commonly used TKI drugs may be given if the first TKI you were given:

- does not work as well as expected
- causes severe side effects.

These TKI drugs include:

Bosutinib (Bosulif®)

You might have bosutinib if other TKIs are causing severe side effects or are not working for you.

Ponatinib (Iclusig®)

Your doctors may prescribe ponatinib if you have leukaemia cells with a gene change (mutation) called T3151. Only a few people with CML have this gene change in their leukaemia cells.

You may also be offered ponatinib if other TKIs have stopped working, or if you had to stop taking them because of side effects.

Asciminib (Scemblix®)

Your doctor may prescribe asciminib if you have all the following:

- chronic phase CML
- 2 or more TKIs stopped working for you or caused severe side effects
- no change in the T3151 gene.

Different drugs are used for different situations. Your haematologist will discuss which is best for you.

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Although the TKIs are similar, they can have different side effects. To make sure the TKI you have is right for you, your haematologist will think about:

- how likely the CML is to respond
- any other health problems you have
- the possible side effects.

If TKIs stop working for you, you may be referred to a specialist CML centre. Your haematologist will talk to you about this if needed.

Before starting TKI treatment, your haematologist will arrange blood tests and heart tests to make sure it is safe for you to have TKIs.

TKI and HIV, hepatitis B and hepatitis C

Before you begin treatment with a TKI, you will have blood tests to check for:

- HIV – a virus that attacks the immune system
- hepatitis B and C – these are liver infections.

TKI treatment can affect your immune system and make hepatitis infections active again. Your doctor or nurse will talk to you about this before your blood tests.

TKI and heart tests

TKIs can affect how your heart works. Before you begin treatment, you have tests to check how well your heart is working. This usually includes an electrocardiogram (ECG) to record the rhythm and electrical activity of your heart.

If you already have heart problems, your haematologist may refer you to a heart specialist before starting treatment.

Side effects of TKIs

The side effects of TKIs are usually mild and treatable. Side effects are often more noticeable when you first start treatment and may get better over time.

If you have severe side effects, your doctor may ask you to stop taking the drug for a few days. After a short break, you may be able to start taking it again without having the same problems. Sometimes, people need to stop treatment with the TKI they are taking because their side effects are too severe. If this happens, you will usually be offered a different TKI drug.

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Sometimes a new side effect can develop many months after starting treatment. Always tell your doctor if you notice any new symptoms or your side effects get worse. Your symptoms may not be related to CML or your treatment, but you should always check with your doctor.

Each TKI has slightly different side effects, so it is best to read specific information about the drug you are having.

Fertility and pregnancy with TKIs

Because TKIs are a newer type of drug, there is limited information about becoming pregnant or getting someone pregnant while taking them. But TKIs are not thought to affect your ability to become pregnant or make someone pregnant. This is called your fertility.

If you might want children in the future, talk to your doctor about this as early as possible. If you can, talk about it before starting treatment. They may refer you to a specialist CML unit or fertility expert. They can talk to you about the possible options for planning your treatment.

Taking a TKI during pregnancy increases the risk of harm to a developing baby. Because of this, you are strongly advised to use contraception while being treated with a TKI.

If you think you may have become pregnant while taking a TKI, tell your doctor as soon as possible. This is because the highest risk to the baby is during the first few weeks of the pregnancy.

Your doctor can talk to you about the possible options for planning your treatment and controlling the CML. They will aim to make things as safe as possible for you and your baby.

Monitoring response to TKI treatment

When you first start treatment with a tyrosine kinase inhibitor (TKI), you will be monitored by your healthcare team every 1 to 2 weeks.

At these check-ups, your doctor will:

- ask about your general health
- ask about any new symptoms or side effects of treatment
- do a blood test to check the numbers of blood cells – this is called a full blood count (FBC).

Your doctor will do a polymerase chain reaction (PCR) blood test to check for leukaemia cells every 3 months to start with. If you have a good response to TKI treatment, this may be done less often.

Sometimes they may take a bone marrow sample. Your doctor can tell you how often you might need this. If you were diagnosed in the [chronic phase], you will not usually need another bone marrow sample taken. This is because PCR testing is very effective in checking how much leukaemia is present.

These test results help your doctors know how well the treatment is working. They will also check for any side effects and make any changes if needed.

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As time goes on, you will not need to meet with your doctors as often. Eventually, you may only need a check-up every 3 to 6 months.

Levels of response to treatment in CML

The aim of treatment is to put the CML into remission. This means there are low levels or no signs of CML in your blood during a standard blood test. It does not mean the leukaemia has completely gone. You will need to keep taking treatment to keep the leukaemia in remission. Because there are still leukaemia cells, your doctors may use the word response instead of remission.

There are different levels of response. These are based on the results of different tests that look at how leukaemia cells respond to treatment.

Your doctors will regularly monitor your response to treatment. This is to check how well it is working for you. We explain the different levels of response below.

Haematological response

This is the first level of response to treatment. It is measured with a full blood count (FBC).

When you first develop CML, the number of white blood cells in your blood is usually high. If there is a complete haematological response, it means:

- your FBC has gone back to normal
- the doctors cannot see any leukaemia cells
- if your spleen was large before starting treatment, it has gone back to a normal size

Most people get a complete haematological response within 3 months of starting a TKI. Although your blood counts are normal, there may still be leukaemia cells that cannot be detected by an FBC.

Cytogenetic response

This refers to the amount of Philadelphia chromosome in the bone marrow and blood. It involves having a bone marrow sample taken. As your treatment starts working, the number of Philadelphia chromosome-positive (Ph+) cells in your bone marrow reduces.

A complete cytogenetic response means there are no Ph+ cells in the bone marrow sample. It takes longer to get a cytogenetic response than a haematological response.

Very few people need another bone marrow sample. You only have this done if your doctor thinks it would be useful in your situation.

Molecular response

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Molecular response is the standard way of measuring your response to treatment. It uses the PCR test, which is very sensitive for finding leukaemia cells. This is important as there may only be 1 leukaemia cell among many thousands of normal blood cells.

There are different levels of molecular response:

- **MR2** means there is less than 1 leukaemia cell in every 100 white blood cells (less than 1%). This response is usually equal to a complete cytogenetic response.
- **MR3 or major molecular response (MMR)** means there is less than 1 leukaemia cell in every 1,000 white blood cells (less than 0.1%).
- **MR4 or deep molecular response (DMR)** means there is less than 1 leukaemia cell in every 10,000 white blood cells (less than 0.01%).
- **MR4.5** means there is less than 1 leukaemia cell in every 32,500 white blood cells (less than 0.003%).
- **MR5** means there is less than 1 leukaemia cell in every 100,000 white blood cells (less than 0.001%).

Continuing with treatment for CML

You will need to keep taking the TKI for as long as it is controlling the leukaemia. This is important, even if your PCR tests show very low levels of leukaemia.

Regularly missing a dose of TKI can affect how well the CML responds to treatment. Research has shown that missing as few as 3 doses a month lowers your chances of getting the best response to treatment.

The following tips may help you remember to take your treatment every day:

- Take your tablets or capsules at the same time each day.
- Set a daily reminder on your mobile phone.
- Put your tablets or capsules in a place where you will notice them every day – but keep them out of sight and reach of children.
- Mark off each dose you take on a calendar.
- Keep a supply of tablets or capsules with you when you travel and take your medicine in your carry-on luggage when you fly.

Your prescriptions will be organised through the hospital, so you may have to go there to collect the treatment each time you need more. If it is difficult for you to get to the hospital, tell your doctor, nurse or pharmacist.

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Treatment free remission in chronic myeloid leukaemia (CML)

Clinical trials have shown that some people who have been taking TKI drugs for a few years and had a good response may be able to stop TKI treatment. This is called treatment-free remission.

If your TKI treatment is stopped, you will be closely monitored so that treatment can be started again if needed. Your doctor can tell you more about this.

About our information

This information has been written, revised and edited by Macmillan Cancer Support's Cancer Information Development team. It has been reviewed by expert medical and health professionals and people living with cancer.

References

Below is a sample of the sources used in our chronic myeloid leukaemia (CML) information. If you would like more information about the sources we use, please contact us at informationproductionteam@macmillan.org.uk

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Our cancer information meets the PIF TICK quality mark.

This means it is easy to use, up-to-date and based on the latest evidence. Learn more about how we produce our information.

The language we use

We want everyone affected by cancer to feel our information is written for them.

We want our information to be as clear as possible. To do this, we try to:

- use plain English
- explain medical words
- use short sentences
- use illustrations to explain text
- structure the information clearly
- make sure important points are clear.

We use gender-inclusive language and talk to our readers as ‘you’ so that everyone feels included. Where clinically necessary we use the terms ‘men’ and ‘women’ or ‘male’ and ‘female’. For example, we do so when talking about parts of the body or mentioning statistics or research about who is affected.

You can read more about how we produce our information here.

How we can help

Macmillan Support Line

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The Macmillan Support Line is a free and confidential service for people living with and affected by cancer. If you need to talk, we'll listen.

Call us free on 0808 808 0000

8am to 8pm, 7 days a week. You can talk to our cancer specialists about anything.

Email the Macmillan Support Line

Email the Macmillan Support Team if you have any questions.

Chat online

Available 8am to 7.45pm, 7 days a week. For confidential, real-time support if you don’t want to call.

Online Community

Chat online anonymously to others who understand what you are going through. Our community is available 24/7 and has dedicated forums where you can get advice and ask our experts.

Find cancer support services in your area

What's going on near you? Find out about support groups, where to get information and how to get involved with Macmillan where you live.

Get help

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